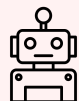


Detection: Find uncertainty in denoising steps

Diffusion-Style Generation: Model iteratively denoises [MASK] tokens



Model is Over-Confident...: But can still be factually wrong



Final Answer:

The city of USA

High Confidence

P(model output) = 0.95

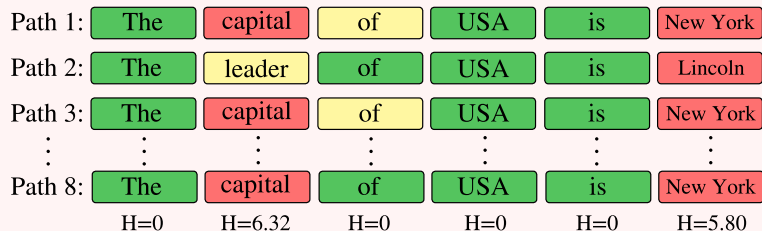
✗ Factually Wrong

GT: The President of USA
is Washington

OSCAR Hallucination Detector: Multi-Path Uncertainty & Entropy Analysis

[MASK] [MASK] [MASK] [MASK] [MASK] [MASK] [MASK]

Build (n = 8) Parallel Trajectories



Mitigation: Targeted entropy-guided remasking

Hallucination Mitigation: Remask only Flagged Tokens

Flagged: [MASK] of [MASK]

Iterative Re-denoising (k steps): Generate n paths per step, track entropy



Majority Voting by Low Entropy: Select token that is

- ✓ Majority Vote by Low Entropy
- ✓ Low entropy (high agreement)

Selected Replacement:

President Washington

✓ **Hallucinations Corrected**

Only uncertain tokens are re-generated -
grounded tokens remain untouched.